Mathematics for Forestry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Mathematics for Forestry (A06805)

Short Title: Mathematics for Forestry
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author ADALYWALSH
Credits 5 **Level:** Introductory

Description of Module / Aims

This is an introductory course in mathematics for foresters. It is assumed that the student has knowledge of basic arithmetic and algebra for this module. The course material is made relevant to practicing foresters where it introduces the geometry applied to angles, length, area, volume and density. Functions and graph interpretation are also introduced.

Programmes

MTHS-0003	BSc in Forestry (WD_SFORS_D)	1 / 1 / M

Indicative Content

- Use of scientific calculator
- Precision, accuracy and SI units
- Geometry of regular and irregular shapes
- Mensuration: areas; volume and density
- Trigonometry: degrees; radians; angles of elevation/depression; angles of elevation/depression; sine rule and cosine rule; area
- Use technology to graph linear, polynomial and growth functions.
- Linear functions: Solution of linear simultaneous equations
- Polynomial functions: solution of quadratic equations and turning points
- Growth functions: indices; logarithms; solution of exponential and logarithm equations

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use appropriate international units (SI units) for the forestry industry.
2. Compute arithmetic and algebraic calculations.
3. Apply the geometry involved in the computation of length, area, volume and density where relevant to the forestry industry.
4. Use and apply trigonometric functions in the context of land surveying.
5. Construct the graphs of linear, polynomial and growth functions using appropriate technology (e.g. excel).
6. Interpret the graphs of linear, polynomial and growth functions.
7. Interpret the results of applied mathematical problems.

Learning and Teaching Methods

- Lectures, In-class exercises, discussion
- Self-directed learning is emphasized.
- Student learning is supported with notes, online material, use of technology and feedback

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,6,7
Continuous Assessment	30%	
Practical	30%	5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of key concepts of mensuration, trigonometry and linear and non-linear models. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments.

40%-49%: Demonstrate a limited knowledge of mensuration, trigonometry and linear and non-linear models. Show some technical ability/skill in carrying out assessments. Attempt made to meet objectives and carry out independent study.

50%-59%: Demonstrate satisfactory general knowledge of mensuration, trigonometry and linear and non-linear models. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study.

60%-69%: Demonstrate sound knowledge of mensuration, trigonometry and linear and non-linear models. Show ability to analyse problems and logically interpret results.

70%-100%: Demonstrate the ability to analyse and design solutions to applied mathematical problems to a high standard, for a range of complex or unseen problems.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Croft, A. and R. Davison. _Foundations Maths (Essential Maths for Students)_. 3rd ed. US: Addison, Wesley Longman, 1997.

Requested Resources

- Room Type: Computer Lab